

Asif Azad Remo

Civil Engineer | Disaster Risk Reduction, Shelter & GIS

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PROFESSIONAL SUMMARY

Civil Engineer (B.Sc., CGPA 3.72/4.00) completing an M.Eng. in Disaster Engineering & Management at CUET, specialising in disaster risk reduction (DRR), flood hazard and vulnerability mapping, and community-based risk reduction. Trained in construction quality control (RCC building works) and GIS, with field experience in community surveys, focus group discussions, and site supervision. Fluent in Bengali, English, and the Chittagonian dialect — a strong fit for CCCM, shelter, and DRR delivery in the Cox's Bazar and Rohingya-camp context.

EDUCATION

M.Eng., Disaster Engineering & Management — CUET, Chattogram Jan 2024 – Present

Thesis: Assessment of Flash Flood Risks in Chattogram City & Risk Reduction through a Community-Based Approach (CBRR)

CGPA: Result not out yet.

B.Sc. in Civil Engineering — PCIU, Chattogram Jan 2019 – Jan 2023

CGPA: 3.72 / 4.00 (159 credit hours)

Thesis: Water Quality Assessment of Halda River at Various Depths

COURSEWORK

M.Eng. — Disaster Engineering & Management, CUET: Challenges in Disaster & Environmental Engineering; Disaster & Emergency Management; Research Methodology & Tools in Disaster Management; Flood Control & Drainage Engineering; Climate Change: Mitigation & Adaptation Strategies; Environmental Impact Assessment; Environmental Management; Environmental Sanitation; Advanced Environmental Bioremediation; Waste Reduction, Water & Wastewater Treatment; Challenges of Concrete Construction for Extreme Conditions; Thesis Project

B.Sc. — Civil Engineering, PCIU: Environmental Engineering; Structural Analysis & Design; Design of Steel Structure; Prestressed Concrete; Geotechnical Engineering; Hydrology; Strength of Materials; Engineering Mechanics; Surveying

RELEVANT EXPERIENCE

Thesis Researcher — Flash Flood Risk & Community-Based Risk Reduction (CBRR) 2025

M.Eng. Research, Chittagong University of Engineering & Technology (CUET), Chattogram

- Conducted GIS-based flood-hazard mapping of Chattogram City (ArcGIS / QGIS) using rainfall, elevation, drainage, slope, DEM, and land-use data to model flood depth and extent
- Developed a composite vulnerability index (socio-economic exposure, sensitivity, adaptive capacity) and overlaid it with hazard maps to produce ward-level risk maps
- Designed and ran community surveys (KoboToolbox), key informant interviews, and focus group discussions across 5+ high-risk wards, capturing local coping mechanisms and gender-specific flood impacts
- Built a Community-Based Risk Reduction (CBRR) framework — early warning, drainage maintenance, flood-proofing, and evacuation — comparable to Community-Led Projects (CLP) in humanitarian DRR
- Coordinated stakeholder validation workshops with local government, NGOs, and community representatives, and drafted a policy brief for municipal disaster planning

Engineering Intern — Site Works & Quality Control Feb – Mar 2023

CPDL, Chattogram

- Supervised and coordinated a 15-person team on active residential and commercial sites, delegating daily tasks and reporting progress to senior engineers
- Carried out site inspections and enforced safety, PPE, and regulatory compliance
- Supported planning and execution to keep works on quality, budget, and schedule
- Maintained site documentation, records, and progress reports

Field Data Collector — Urban Water-Logging (Flood) Vulnerability Assessment Sep 2021

Chittagong University of Engineering & Technology (CUET), Chattogram

- Collected and organised field data to assess urban flood / water-logging vulnerability across affected areas
- Supported the research team in identifying key hazard-contributing factors
- Documented and analysed findings to support project outcomes

Supervisor: Md. Mehedi Hassan Masum, CUET — mehedi.ce@cuet.ac.bd

Undergraduate Researcher — Water Quality of the Halda River Jan – Apr 2020

Port City International University (PCIU), Chattogram

- Led an independent thesis assessing water quality across river depths using field and laboratory methods
- Analysed environmental parameters to evaluate river health and ecosystem sustainability
- Presented findings at the 7th International Conference on Civil Engineering for Sustainable Development (ICCESD)

Supervisor: Md. Jahedul Alam, PCIU — jahed150@gmail.com

CERTIFICATIONS & TRAINING

- **GIS and Its Applications** (80 hours) — EDGE, in collaboration with CUET, 2024. Spatial data models & coordinate systems, digitisation, geodatabase management, spatial analysis (buffer, overlay, proximity, network), terrain & surface analysis, cartography, and remote-sensing integration — with applications in disaster management (flood, cyclone, earthquake risk mapping).
- **Quality Control of RCC Building Construction** (40 classes / 40 hours) — Qlearn; Instructor: Prof. Dr. Jahangir Alam, Dept. of Civil Engineering, BUET. Site survey, soil-test supervision, drawing interpretation, **estimating**, site layout, construction-material QC, and concrete-casting & reinforcement quality control across footing, pile cap, column, beam, slab, stair, shear wall, and lintel; precast & cast-in-situ piling; coordination with owners, foremen, and labour contractors.
- **Run Your First Python Code** — OSTAD, 2024.

TECHNICAL SKILLS

Software & Tools: ArcGIS Pro, ArcMap, QGIS, AutoCAD, UTM Geo Map, MS Excel & MS Office, Google Workspace, KoboToolbox, SPSS, R, Python, Mendeley, VS Code

Analytical / GIS: GIS mapping & spatial analysis, terrain & vulnerability analysis, statistical analysis, environmental-standards comparison (BECR 2023, WHO, FAO), research documentation & reporting

Field & Laboratory: Water sampling (Van Dorn) & preservation, pH, EC, TDS, DO & COD (titration), spectrophotometry, turbidity, salinity assessment

LANGUAGES

- **Bengali:** Native / Advanced
- **English:** Advanced
- **Chittagonian:** Native / Fluent

PUBLICATION

Azad Remo, A., Rashid, R., Das, B., & Hossan, M.S. (2024). *Water Quality Assessment of Halda River at Various Depths*. Proceedings of the 7th International Conference on Civil Engineering for Sustainable Development (ICCESD). [View on ResearchGate](#).

ACHIEVEMENTS

- University Merit Scholarship (2nd–4th year) and National Education Board Scholarship (JSC)
- Participant, Inter-University Mechanics Olympiad 2022, PCIU

REFERENCES

Dr. Md. Abul Hasan — Head & Professor, Dept. of Civil Engineering, CUET | headdem@cuet.ac.bd | +880 1609 495478

Md. Mehedi Hassan Masum — Lecturer, Dept. of CRHLSR, CUET | mehedi.ce.cuet@gmail.com | +880 1683 688126